2.4 $\pi^2/6$

WORKSHEET

The question

STEP III 2018, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If a polynomial $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has roots $r_1, \ldots, r_n$, what is $r_1 + \cdots + r_n$ in terms of the coefficients?

STAGE 1 De Moivre

Covers: Part (i)

Ask yourself Write $\cot\theta + \mathrm{i}$ as a single fraction over $\sin\theta$. What do you notice about the numerator?

Pause and try Raise both $\cot\theta + \mathrm{i}$ and $\cot\theta - \mathrm{i}$ to the power $2n+1$ with De Moivre's theorem, and subtract.

STAGE 2 The polynomial

Covers: Part (i)

Ask yourself Now expand $(c+\mathrm{i})^{2n+1} - (c-\mathrm{i})^{2n+1}$ with the binomial theorem, where $c = \cot\theta$. Which terms cancel, and which double?

Pause and try For which θ is $\sin(2n+1)\theta = 0$ but $\sin\theta \neq 0$? Use these values to find n different roots of the polynomial in $x = \cot^2\theta$. Why are they all of its roots?

STAGE 3 Adding the roots

Covers: Part (ii)

Pause and try If a polynomial $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has roots $r_1, \ldots, r_n$, then $r_1 + \cdots + r_n = -a_{n-1}/a_n$. Read off the two coefficients from part (i).

STAGE 4 The squeeze

Covers: Part (iii)

Ask yourself Start from $0 < \sin\theta < \theta < \tan\theta$ for $0 < \theta < \frac{\pi}{2}$. Take reciprocals and square. How is $\frac{1}{\sin^2\theta}$ related to $\cot^2\theta$?

Pause and try Put $\theta = \frac{m\pi}{2n+1}$ and add over $m = 1, \ldots, n$. Part (ii) gives the two outer sums. What do the bounds tend to as $n \to \infty$?

What's really going on

Euler's bold argument

Euler's own 1734 argument was not rigorous by modern standards, but it was dazzling. A polynomial with roots $\pm a, \pm b, \pm c, \ldots$ and value 1 at 0 factorises as $(1 - x^2/a^2)(1 - x^2/b^2)\cdots$. The function $\frac{\sin x}{x}$ has value 1 at 0 and roots $\pm\pi, \pm 2\pi, \pm 3\pi, \ldots$, so, treating it as an "infinite polynomial", Euler wrote

$$\frac{\sin x}{x} = \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\cdots.$$

The coefficient of x^2 on the right is $-\frac{1}{\pi^2}\left(1 + \frac{1}{4} + \frac{1}{9} + \cdots\right)$, while from the Maclaurin series $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \cdots$ it is $-\frac{1}{6}$. Comparing gives $\pi^2/6$. Our STEP proof is the finite, rigorous version of the same idea: a polynomial whose roots are tied to multiples of π, and its second coefficient.

Beyond reciprocal squares

Euler went on to find $\sum 1/m^4 = \pi^4/90$, $\sum 1/m^6 = \pi^6/945$, and in general $\sum 1/m^{2k}$ is a rational multiple of π^{2k}, involving the *Bernoulli numbers*. The odd powers are a different story: as we saw in Section 2.3, nobody knows a formula for $\sum 1/m^3$. These sums are the values $\zeta(2), \zeta(4), \ldots$ of the Riemann zeta function, whose behaviour for complex arguments controls the distribution of the primes (Section 1.2).

Where you will see it again

At university you will meet at least three more proofs of $\zeta(2) = \pi^2/6$: one from *Fourier series* (writing a function as a sum of sines and cosines), one from a double integral, and one from complex analysis. And the number turns up in probability: the chance that two randomly chosen whole numbers have no common factor is $6/\pi^2 \approx 0.61$. (A prime p divides both with probability $1/p^2$, so the chance that no prime divides both is $\prod_p (1 - 1/p^2)$, and Euler showed this product equals $1/\zeta(2)$.) A different appearance of π in a probability limit is the subject of Section 12.23.

Try this next

1. **The 1992 version.** Try STEP III 1992, Question 8 (step.maths.org/questions/92-s3-q8). It uses a polynomial like the one in this section to find exact sums of squared tangents and cosecants. (*Hint:* if you know a polynomial whose roots are r_i, what polynomial has roots $1/r_i$?)

2. **Odd squares.** Deduce from $\sum 1/m^2 = \pi^2/6$ that $1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \cdots = \frac{\pi^2}{8}$.

3. **Fourth powers.** HARDER Use $\sum r_i^2 = (\sum r_i)^2 - 2\sum_{i<j} r_i r_j$ and the coefficients of the polynomial to find $\sum_{m=1}^{n} \cot^4\left(\frac{m\pi}{2n+1}\right)$, and adapt the squeeze to prove $\sum 1/m^4 = \pi^4/90$.

4. **How fast?** Using the lower bound in part (iii), show that $\frac{\pi^2}{6} - \sum_{m=1}^{n} \frac{1}{m^2} < \frac{\pi^2}{4n}$. How large an n does this guarantee is enough for the partial sum to be within 0.001 of $\pi^2/6$? Compare with the true error, which is close to $1/n$.

Further reading

- **William Dunham, *Euler: The Master of Us All*** (Mathematical Association of America). Chapter 3 tells the story of the Basel problem with the mathematics in full.
- **Aigner & Ziegler, *Proofs from THE BOOK***, Chapter 9, "Three times $\pi^2/6$". Includes the proof in this section.
- **Paul Nahin, *Dr. Euler's Fabulous Formula*** (Princeton University Press). Complex numbers, Fourier series and sums like this one.

The full guided solution is in Section 2.4:
https://beyondstep.keithmcnulty.org/ch02/basel.html